



Can scintillation detectors with low spectral resolution accurately determine radionuclides content of building materials?

K. Kovler^{a,*}, Z. Prilutskiy^a, S. Antropov^b, N. Antropova^b, V. Bozhko^b, Z.B. Alfassi^c, N. Lavi^c

^a National Building Research Institute—Faculty of Civil and Environmental Engineering, Technion—Israel Institute of Technology, Haifa 32000, Israel

^b “Amplituda” Research Center, POB 120, Moscow 124460, Russia

^c Department of Nuclear Engineering, Ben-Gurion University, Beer Sheva 84105, Israel

HIGHLIGHTS

- NORM activity concentrations in various materials were measured by NaI(Tl) and HPGe spectrometers.
- Synthetic compositions and building materials with different NORM concentration were studied.
- Densities of the tested samples varied in a wide range (from 860 up to 2,410 kg/m³).
- The results by the NaI(Tl) system were close to those obtained with the HPGe spectrometer.
- Special software compensates for lower spectral resolution of NaI(Tl) detectors.

ARTICLE INFO

Article history:

Received 21 June 2012

Received in revised form

19 February 2013

Accepted 27 February 2013

Available online 14 March 2013

Keywords:

Natural radionuclides

Building materials

Scintillation spectrometry

High-purity germanium detectors

ABSTRACT

The current paper makes an attempt to check whether the scintillation NaI(Tl) detectors, in spite of their poor energy resolution, can determine accurately the content of NORM in building materials. The activity concentrations of natural radionuclides were measured using two types of detectors: (a) NaI(Tl) spectrometer equipped with the special software based on the matrix method of least squares, and (b) high-purity germanium spectrometer. Synthetic compositions with activity concentrations varying in a wide range, from 1/5 to 5 times median activity concentrations of the natural radionuclides available in the earth crust and the samples of popular building materials, such as concrete, pumice and gypsum, were tested, while the density of the tested samples changed in a wide range (from 860 up to 2,410 kg/m³). The results obtained in the NaI(Tl) system were similar to those obtained with the HPGe spectrometer, mostly within the uncertainty range. This comparison shows that scintillation spectrometers equipped with a special software aimed to compensate for the lower spectral resolution of NaI(Tl) detectors can be successfully used for the radiation control of mass construction products.

© 2013 Elsevier Ltd. All rights reserved.

1. Introduction

Most building materials of terrestrial origin contain small amounts of Naturally Occurring Radioactive Materials (NORM), mainly radionuclides from the ²²⁶Ra and ²³²Th decay chains and the radioactive isotope of potassium, ⁴⁰K. The ²²⁶Ra series is often called ²³⁸U series in the literature. ²³⁵U is of minor significance due to its low abundance in comparison with ²³⁸U.

The median values of the activity concentrations of ²²⁶Ra, ²³²Th and ⁴⁰K in the earth's crust are 35, 30 and 400 Bq/kg, respectively (UNSCEAR, 2000).

An inhabitant living in an apartment block made of concrete with average activity concentrations (40, 30 Bq/kg and 400 Bq/kg for ²²⁶Ra, ²³²Th and ⁴⁰K, respectively) receives an annual effective

dose of about 0.25 mSv (excess to the external gamma radiation dose received outdoors) (RP-112, 1999). It is known that enhanced or elevated levels of natural radionuclides in building materials may cause doses in the order of several mSv/year. The purpose of setting controls on the radioactivity of building materials is to limit the radiation exposure due to materials with enhanced or elevated levels of natural radionuclides.

The doses to the members of the public should be kept as low as reasonably achievable. However, since small exposures from building materials are ubiquitous, controls should be based on exposure levels which are above typical levels of exposures and their normal variations.

The following activity concentration index (*I*) is often derived for identifying whether a dose criterion is met:

$$I = \frac{A_{\text{Ra}}}{300} + \frac{A_{\text{Th}}}{200} + \frac{A_{\text{K}}}{3000} \leq 1 \quad (1)$$

* Corresponding author. Tel.: +972 4 8292971; fax: +972 4 8295697.
E-mail address: cvrkost@technion.ac.il (K. Kovler).

Table 1

Dose criterion recommended by EC (RP-112, 1999).

Dose criterion	0.3 mSv yr ⁻¹	1.0 mSv yr ⁻¹
Materials used in bulk amounts, e.g. concrete	$I \leq 0.5$	$I \leq 1$
Superficial and other materials with restricted use: tiles, boards, etc.	$I \leq 2$	$I \leq 6$

where A_{Ra} , A_{Th} and A_{K} are activity concentrations (in Bq/kg) of natural radionuclides ^{226}Ra , ^{232}Th and ^{40}K , respectively. According to RP-112 (1999), the activity concentration index I shall not exceed the following values depending on the dose criterion and the way and the amount the material is used in a building (Table 1).

The activity concentrations of natural radionuclides ^{226}Ra , ^{232}Th and ^{40}K are determined by γ -ray spectrometry. The activities of ^{226}Ra and ^{232}Th are usually determined by measuring the activities of their progeny: ^{214}Pb and ^{214}Bi (for ^{226}Ra) and ^{228}Ac , ^{212}Pb , ^{212}Bi and ^{208}Tl (for ^{232}Th).

The advantages of the gamma-ray spectrometry are: high sensitivity, non-destructive mode of operation, short test duration and simplicity. Most common detectors include sodium iodide doped with thallium, or NaI(Tl), scintillation counters and high-purity germanium (HPGe) detectors. NaI(Tl) detectors are simpler and cheaper, but have poor energy resolution. For this reason they are often considered as not suitable for the identification of complicated mixtures of γ -ray-producing materials and quantitative determination of their radionuclide composition.

The current paper makes an attempt to check whether the scintillation NaI(Tl) detectors, in spite of their poor energy resolution, can determine accurately the content of NORM in building materials.

2. Semiconductor (HPGe) gamma spectrometers

Germanium detectors have better energy resolution as compared to that of NaI(Tl) detectors and are better suited to the assignment of resolving various gamma-ray spectra. HPGe gamma spectrometers provide information about the isotopic content of materials. The germanium detectors used by the IAEA range in size from small planar types to large (80–90 cm³) coaxial detectors. Semiconductors must exhibit rapid and effective transport of electrons and holes, and they must be capable of operation in a mode in which surface electrodes are used to apply a strong electric field without an unacceptably large current flow (Gilmore, 2008).

One of the disadvantages of the HPGe detector is that it can only function as a spectrometer if cooled to liquid-nitrogen temperatures, otherwise electrons can be thermally excited into the conduction-band and so generate a high level of noise. This means that an HPGe detector is neither compact nor rugged.

Nowadays electric cooling systems have become available, mitigating this disadvantage with minor effect on detector performance characteristics.

The second disadvantage is that in order to provide a stopping power equivalent to a commonly available size of scintillation spectrometer, the germanium crystal becomes very expensive to fabricate. As a consequence, HPGe detectors are significantly more expensive than NaI(Tl) detectors (IAEA, 2011).

3. Scintillation (NaI(Tl)) gamma spectrometers

NaI(Tl) detectors are commonly used to identify and measure activities of low-level radioactive sources. They have high

Table 2

Comparison of HPGe and NaI(Tl) detectors, by Gilmore (2008).

HPGe	NaI(Tl)
Operated at liquid nitrogen temperature (77 K)	Cheaper (× 10)
Insensitive to temperature	More efficient (× 10)
Insensitive to bias voltage	Large volumes available
Good energy resolution	Operated at room temperatures
	Sensitive to temperature
	Sensitive to anode voltage
	Poor energy resolution

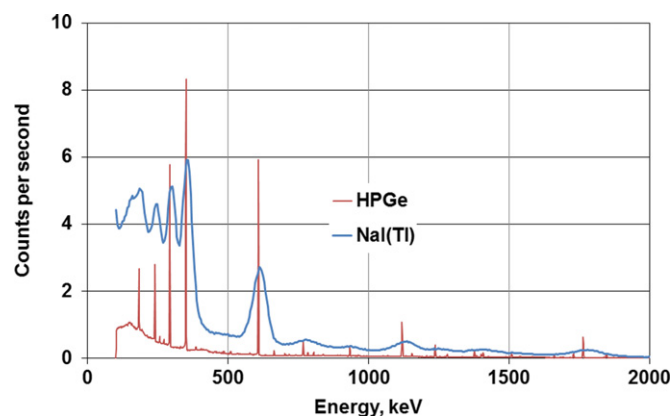


Fig. 1. Spectra of the calibrated sources of ^{226}Ra calibrated source measured by means of scintillation NaI(Tl) and semiconductor HPGe γ -ray detectors.

detection efficiency and operate at room temperature. One of the most important parameters in the calculation of the gamma activity of environmental radioactive sources is detection efficiency. It is usually determined by using calibrated standard sources.

Gamma detection techniques are widely used in gamma spectroscopy for nuclear physics, medical radiography, neutron activation analyses, and study of cosmic rays. However, the ability of scintillators to distinguish between γ -rays of different energies is relatively poor, and the detector types have the worst energy resolution, although NaI(Tl) detectors have the highest efficiency and the lowest minimum detectable activity (Perez-Andujar and Pibida, 2004). Table 2 compares the differences between NaI(Tl) and HPGe detectors.

Fig. 1 shows energy spectra of the calibrated ^{226}Ra source measured by means of scintillation NaI(Tl) and semiconductor HPGe detectors. It can be seen that the peaks obtained by the NaI(Tl) detector are rather wide, while the neighboring peaks overlap, which may make difficult the source identification, especially in materials having a complex radionuclide composition.

Because of the poor resolution scintillation detectors are often considered as not suitable for the identification of complicated mixtures of γ -ray-producing materials and quantitative determination of their radionuclide composition. Indeed, HPGe detectors provide significantly improved energy resolution in comparison to NaI(Tl) detectors; however cryogenic temperatures are vital to their operation, which makes the maintenance of the spectrometric system more complicated and costly. Relatively high costs of HPGe detectors and their maintenance, in addition to high skills required from the personnel, limit the use of HPGe detectors in special research laboratories (usually national research centers or universities), while their use in the mass control of products (such as building materials) is problematic.

At the same time, the practice shows that NaI(Tl)-based detectors can be successfully used for quantitative determination

of activity concentrations of the mixtures of few and known radionuclide composition, such as those containing natural radionuclides (NORM) only. The present paper deals with building materials of mineral origin, so we are looking at only three candidates for what is essentially an identification of the ^{40}K , and the ^{232}Th and ^{238}U series. In some countries, like Finland, Ukraine, Belarus and Russia ^{137}Cs activity should be controlled as well (Guide, 2005; GOST 30108, 1994). These four candidates are a constraint that limits possible radionuclide composition.

At the same time, not every scintillation spectrometer is suitable for such purpose. For performing an accurate quantitative analysis, the NaI(Tl)-based system should have a special software, capable of distinguishing between these four different radionuclides in the mixture (NORM + ^{137}Cs) and accurately determining their activity concentrations.

In many applications, such as the mass control of radioactive impurities in building materials, it is not practical to use a cooled high-purity germanium spectrometer. The improved efficiency, for the same size detector, and the lower cost of a NaI(Tl) detector must be traded-off against the better resolution of HPGe detector.

4. How to compensate for the lower spectral resolution of scintillation spectrometers?

The popular method to compensate for the lower spectral resolution of NaI(Tl) detectors is to apply spectral deconvolution to the raw energy-loss data collected by the spectrometer. Deconvolution is a technique used in spectroscopy and other diverse fields, in which a raw data spectrum obtained with a detection system is deconvolved with a response function representing the response of the detection system to known input signals (Meng and Ramsden, 2000).

A simplified approach to spectral deconvolution was suggested by Rybach (1988). Three counting windows or regions of interest (ROI) in the γ -ray spectrum were identified. They were centered on the three characteristic photo-peaks, at approximately 1.46 MeV (^{40}K), 1.76 MeV (^{214}Bi) and 2.62 MeV (^{208}Tl), corresponding to the ^{40}K , ^{226}Ra and ^{232}Th decay series. The system calibration was done by using three reference materials, obtained from the International Atomic Energy Agency for ^{40}K , ^{226}Ra and ^{232}Th (IAEA, 1987). Rybach (1988) suggested that ROIs should be as wide as 10% of the energy peak of a characteristic nuclide.

Chiozzi et al. (2000) generalized the procedure proposed by Rybach (1988). They assume that the net count rate $\alpha_{i,j}$ in the i^{th} ROI of a calibration standard j (with i and j equal to 1, 2 and 3 denoting the ROIs, and the calibration standards of ^{40}K , ^{226}Ra and ^{232}Th , respectively) is proportional to the activity $A_{n,j}$ of each investigated nuclide n ($n=1, 2$ and 3 for ^{40}K , ^{226}Ra and ^{232}Th , respectively) according to

$$\alpha_{i,j} = \sum_{n=1}^3 e_{i,n} A_{n,j} \quad (2)$$

where $e_{i,n}$ is the counting efficiency in the i^{th} ROI for the nuclide n . The net count rate is given by

$$\alpha_{i,j} = \frac{N_{i,j}}{t_j} - R_{i,b} \quad (3)$$

where t_j is the counting time for the standard j , $N_{i,j}$ is the corresponding number of counts and $R_{i,b}$ is the background count rate in the i^{th} ROI.

Eq. (2) represents a linear system of $(i=3) \times (n=3)$ simultaneous equations that can be univocally solved with respect to the nine counting efficiencies, on the basis of the measured count rates and activity of the standards. Eq. (2) can be also used to obtain the activity of ^{40}K , ^{226}Ra and ^{232}Th of the sample from the

Table 3

Activity concentrations in rock samples measured by γ -ray systems with HPGe and NaI(Tl) detectors and measurement errors (conf. level=95%); the data reported by Chiozzi et al. (2000).

Sample	Detector	Activity concentrations (Bq/kg)		
		^{238}U	^{232}Th	^{40}K
Andesite	HPGe	65 ± 10	52 ± 4	847 ± 56
	NaI(Tl)	49 ± 6	57 ± 4	946 ± 22
Rhyolite	HPGe	194 ± 24	142 ± 4	1305 ± 86
	NaI(Tl)	130 ± 2	147 ± 4	1292 ± 22
Dolomite	HPGe	69 ± 6	4 ± 2	–
	NaI(Tl)	50 ± 1.5	2 ± 1.6	22 ± 111
Granodiorite	HPGe	64 ± 12	50 ± 4	1016 ± 6
	NaI(Tl)	46 ± 4	56 ± 2	881 ± 18

net count rate measured in the three ROIs (α_K , α_{Ra} and α_{Th}) and the counting efficiencies.

The reliability of the measurements by NaI(Tl) detector was tested by Chiozzi et al. (2000) from a comparison with the measurements conducted with HPGe detector, for four selected rock samples in the same Marinelli beakers. The activity measurements were made for ^{238}U , ^{232}Th and ^{40}K .

It can be seen from Table 3 that activities of ^{232}Th are slightly lower in HPGe detector (excluding the dolomite sample, wherein the activities are close to the detection limit), the average difference being 8%. The maximum relative difference occurs for ^{238}U (25–33%).

5. Experimental comparison of spectrum processing methods

The performance of different spectrum processing methods was studied and compared on the same scintillation detector. For the experiments the scintillation spectrometer “Multirad” (Amplituda, Moscow) with a resolution of 7.3% from the peak ^{137}Cs (662 keV) and efficiency ε (662 keV)=0.023 was used. In the spectrometer software various algorithms of the spectrum deconvolution were implemented. Several methods of spectrum processing have been compared experimentally. Some of these methods were similar to those described in the literature review (see the previous section). The methods of spectrum processing have been implemented as Script-programs in the “Progress” software (Antropov et al., 2003; Antropov, 2004). One of the methods used in the comparative analysis of the scintillation and HPGe spectrometers was applied in further to the problem of NORM measurement in building materials. The principles of spectrum deconvolution implemented in the current study are briefly described hereafter.

It is known that gamma emission in building materials consists of natural radionuclide ^{40}K , daughter nuclides of ^{226}Ra (^{214}Pb , ^{214}Bi) and of ^{232}Th (^{228}Ac , ^{212}Pb , ^{212}Bi and ^{208}Tl). If the material does not undergo a chemical separation process, ^{228}Ac is under equilibrium with its daughters, ^{212}Pb , ^{212}Bi and ^{208}Tl , and their activity is the same as that of all the radionuclides belonging to the ^{232}Th series.

Activity of ^{214}Pb and ^{214}Bi in building materials differs from the activity of their parent nuclide ^{226}Ra because of ^{222}Rn emanation. In order to achieve a radioactive equilibrium the samples should be placed in a hermetically closed container prior to the measurements. In a general case, the presence of ^{137}Cs in building materials cannot be neglected, especially after the Chernobyl disaster. Therefore, the measured spectrum should be a sum of 5 components: external background, emission of natural radionuclide ^{40}K , emission of ^{226}Ra and its daughter nuclides (^{214}Pb , ^{214}Bi), emission of ^{232}Th daughter nuclides (^{228}Ac , ^{212}Pb ,

^{212}Bi and ^{208}Tl and man-made ^{137}Cs :

$$S(E) = Bg(E) + A_{Ra}P^{Ra}(E) + A_{Th}P^{Th}(E) + A_KP^K(E) + A_{Cs}P^{Cs}(E) \quad (4)$$

where: Bg is background count rate; A_{Ra} , A_{Th} , A_K and A_{Cs} —activities of $^{214}\text{Pb}(\text{Bi})$, ^{232}Th , ^{40}K and ^{137}Cs ; P^n —detector sensitivity function of the source n .

In solving Eq. (4) various processing methods can be applied: they use different parts of the initial experimental information. One of the methods uses all the points of the measured spectrum, the other one—only those in the vicinity of the given peak, and the third one—integrals for energy intervals. The availability of the method and its precision serve as the main criteria for the final choice of the spectrum processing method with respect to a particular practical problem.

The example of the spectrum of the probe measured by the scintillation spectrometer NaI(Tl) 63×63 mm in the container with Marinelli geometry is presented in Fig. 2.

Precision is characterized by the uncertainty of activity measurement. The uncertainty depends on both statistical reasons and the differences of the specific model, for which the method is calibrated, from the actual measurement conditions.

In order to reduce the statistical component of the measurement uncertainty, the method of spectrum processing should take into account as much experimental information. An ideal situation would be involving the whole spectrum in the processing procedure. From the point of view of reducing the non-statistical uncertainty components it is sometimes advisable to exclude from the processing that part of the information, which depends mostly on the variation of factors, which are difficult to control.

For example, in the energy range above 300 keV, the main effect resulting in self-absorption of gamma radiation in the sample material is Compton scattering. The probability of this effect is proportional to the electron density or density of the sample. In the energy range below 300 keV self-absorption of the radiation is mainly due to the photoelectric effect, the probability of which depends on the density of the sample, and the atomic number of the material, from which the sample is made. In practice, considering the atomic number of the sample material is not always possible. Therefore, it would be better to exclude the range of energies below 300 keV from the processing, despite the fact that this exclusion will lead to an increase in the statistical component of uncertainty.

According to the Gauss–Markov theorem, the results obtained by the Generalized Least Squares (GLS) method have the least standard deviation among all possible spectrum processing methods. The methods described in further differ in the composition of the initial information. These techniques have been

implemented and compared with the GLS method that uses all the points of the measured spectrum in the range 300–2,800 keV.

To assess the statistical component of the measurement uncertainty and compare between the different spectrum processing methods, 15 consequent measurements of the soil sample of low activity (humus), have been conducted. The duration of each measurement was 1 h, while the duration of background spectrum measurement, used by processing procedure, was 14 h. The obtained spectra were analyzed using the following processing methods:

- (1) Decomposition of the spectrum using the GLS method in the energy range 300–2,800 keV; the spectrum is represented as the sum of the background, ^{137}Cs , ^{40}K , ^{226}Ra , ^{232}Th .
- (2) Matrix method for the following 12 energy regions selected in the energy range 300–2,800 keV: 300–400; 400–580; 580–630; 630–720; 720–800; 800–1,030; 1,030–1,400; 1,400–1,580; 1,580–1,860; 1,860–2,250; 2,250–2,400 and 2,400–2,800 keV.
- (3) Matrix method for the 4 energy regions: 600–720; 1,350–1,560; 1,640–1,880; 2,500–2,750 keV. This is the method that is analogous to those described and applied by Rybach (1988) and Chiozzi et al. (2000).
- (4) Determination of activities of ^{40}K , ^{226}Ra , ^{232}Th by calculation of the areas under the peaks 1460, 1,764 and 2,614 keV, using peak approximation by Gaussian of the known width and location lying on the direct base.
- (5) Determination of activities of ^{137}Cs , ^{226}Ra , ^{232}Th by calculation of the areas of the interfering peaks of 662, 609 and 583 keV. The area under the peak was determined by approximation of the part of the spectrum in the interval 520–735 keV by the sum of three Gaussians of the known width and location lying on the direct base.

The mean values and uncertainties (confidence level of 95%) calculated for the obtained results are assembled in Table 4. Uncertainty was calculated as double standard deviation of the 15 measurements made on the same low activity soil sample.

The compared methods used in this experiment were implemented in the “Progress” software. The processing algorithm may differ from the algorithms of other programs, for example by the approximation of the base line under the peak. However, the uncertainty in general (see Table 4) allows evaluating the stability of algorithms to statistical variations of the spectrum processing method.

Table 4 shows that the GLS method is the best method in terms of accuracy. At the same time, it can be noted that the scatter of the experimentally observed results for ^{232}Th , in the case of the matrix method, is slightly lower than that obtained by the GLS method (lines 1 and 2 of Table 4).¹

For the further comparison with the HPGe-detector, we used the matrix processing method with 12 energy intervals, which is implemented as default software in the scintillation gamma spectrometer “Multirad”. In Table 4, this method is the second best in terms of accuracy (statistical component of the

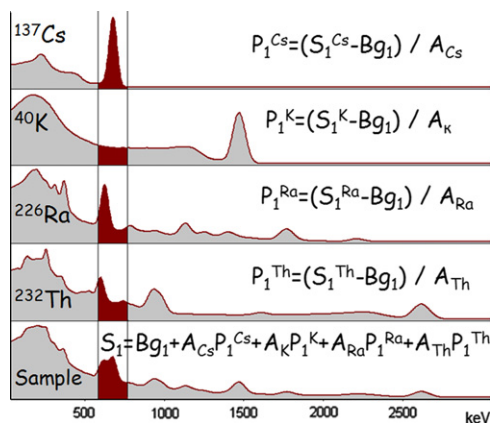


Fig. 2. Spectra of the calibrated sources of ^{226}Ra , ^{232}Th , ^{40}K and ^{137}Cs of the sample containing these nuclides all together.

¹ This finding looks contradicting the Gauss–Markov theorem, according to which the results obtained by the generalized least squares method should have a minimum deviation, but can be attributed to the lower sensitivity of the matrix method to the energy drift of the spectrometer. In fact, the measurements of spectra in our experiment were carried out continuously for 15 hours, without performing energy calibration between the consequent measurements. It has to be emphasized that due to a possible drift of the energy scale the relative changes in the integral over the interval 2,400–2,800 keV are lower than the changes in the counting rate of the individual channels in the vicinity of the 2,614 keV peak.

Table 4

The results of the statistical analysis of 15 measurements of the same soil sample of low activity for the different methods of spectrum processing (A—activity calculated by the software; ΔA —uncertainty at confidence level of 95%).

Spectrum processing method	¹³⁷ Cs		⁴⁰ K		²²⁶ Ra		²³² Th	
	A	ΔA	A	ΔA	A	ΔA	A	ΔA
1. Generalized least squares (300–3000 keV)	2.0	1.1	158.5	16.4	10.6	2.1	11.0	2.5
2. Matrix, 12 intervals	2.0	1.6	153.7	24.8	8.6	2.7	8.6	2.2
3. Matrix, 4 intervals	4.4	1.6	150.6	25.3	4.0	4.6	7.6	3.1
4. Single peak processing; peaks 1460, 1764, 2614 keV.			145.8	27.1	6.4	3.2	7.5	7.5
5. Interfering peak processing; peaks 583, 609, 662 keV.	1.0	1.4			6.6	3.0	5.8	5.3

measurement uncertainty) after the GLS method. Despite this, we have chosen the matrix method with 12 intervals by the following reasons:

- (1) For more than 15 years, this method is widely used in Russian Federation and has been tested on real samples of the environmental probes and building materials. In contrast to the GLM method, it is implemented not only on PC, but also on simple portable devices.
- (2) The matrix method allows the registration of self-absorption in the wide range of density, from 200 to 2,000 kg/m³.
- (3) The borders of energy intervals of the matrix method are chosen so, that the shift of the energy scale would be the least factor affecting the readings. This should reduce the non-statistical components of the measurement error, associated with the temperature drift of the light output of the crystal, which is inevitable when measuring samples of building materials having a temperature significantly different from the room temperature.

To solve Eq. (4), matrix method uses 12 energy intervals in the energy range 300–2800 keV. The intervals are: 300–400; 400–580; 580–630; 630–720; 720–800; 800–1,030; 1,030–1,400; 1,400–1,580; 1,580–1,860; 1,860–2,250; 2,250–2,400 and 2,400–2,800 keV.

According to the additivity principle, the spectrum of the sample, which consists of various components in each energy interval, is the sum of the spectra registered separately from each source in identical conditions:

$$S_i = Bg_i + A_{Ra}P_i^{Ra} + A_{Th}P_i^{Th} + A_KP_i^K + A_{Cs}P_i^{Cs} \quad (5)$$

where P_i^X is a probability that the detector counted 1 impulse from the disintegration of the given radionuclide X within an energy interval i .

The lowest border of 300 keV was chosen so, that the self-absorption of γ -emission by the measured material would be dependent on its density and not be too much influenced by the atomic number. The borders of the intervals were chosen in a way that one or several peaks of the certain nuclide would be located within each energy interval.

In order to consider the dependence between the self-absorption and sample substance the detector sensitivity P_i^X is approximated by the expression:

$$P_i^X = \frac{P_{0i}^X}{\mu_i^X \cdot M} \cdot [1 - \exp(-\mu_i^X \cdot M)] \quad (6)$$

where: M is the mass of the sample, which is proportional to density for the standard sample geometry; i , interval number ($i=1, 2, \dots, 12$); $X=^{226}\text{Ra}$, ^{232}Th ; ^{137}Cs and ^{40}K , P_{0i}^X and μ_i^X , constant coefficients.

The values of coefficients P_{0i}^X are determined by the calibrated sources having a density of 1,000 kg/m³. The coefficients μ

depend on the measurement geometry only, and can be either positive or negative depending on the energy interval. These coefficients are determined with the help of the sources with densities in the range between 200 and 2,000 kg/m³.

The number of equations should exceed the number of variables in the system of Eq. (5), in order to be able using the least squares method.

6. NaI(Tl) vs. HPGe detectors: experimental program of the measurements

The goals of the experimental study were to compare the measurement results obtained with two different detectors, NaI(Tl) and HPGe, and their accuracy.

High-resolution gamma-ray spectrometry system consisted of a 19% efficiency PopTop-type HPGe coaxial detector (manufactured by Ortec) with 2.4 keV resolution at Co-60 1.332 MeV photons, shielded by 50-mm Pb casemate. The efficiency calibration of the gamma spectrometry systems was performed with the radionuclide specific efficiency method in order to avoid any uncertainty in gamma ray intensities as well as the influence of coincidence summation and self-absorption effects of the emitting gamma photons. A set of high quality certified reference materials (IAEA) was used, with densities similar to the building materials measured after pulverization. Cylindrical geometry was used assuming that the radioactivity is homogeneously distributed in the measuring samples. The measurement duration was up to 270,000 s. The detector was installed in a well consisting of 50-mm thick lead, to shield the measuring station against background radioactivity. Standard nuclear electronics were used and the spectra were stored for analysis in 1,024 channels.

Background spectra were also collected for the same period of time. The net sample count rate at each energy peak was obtained after subtraction of the corresponding background rate. The latter was found to be at least 3 times lower than the sample count rate, depending on the peak under study.

The scintillation spectrometer “Multirad” (Amplituda) with NaI(Tl) 63 × 63 mm crystal detector and 50-mm lead protection casemate was used. This spectrometer model is widely used in Russian Federation for the radiation control of mass production in agriculture and construction.

In order to study the influence of the different ratios between natural radionuclides, 8 synthetic mixtures containing the three nuclides, ^{226}Ra , ^{232}Th and ^{40}K , were prepared at two activity concentrations levels, “high” and “low”. These levels were chosen approximately 5 times lower and higher than the median values of the activity concentrations of ^{226}Ra , ^{232}Th and ^{40}K in the earth's crust, i.e. 35, 30 and 400 Bq/kg, respectively (UNSCEAR, 2000). The “high” and “low” levels of activity concentrations are shown in Table 5 and marked as “+” and “−”, respectively.

7. Measurement uncertainty

The efficiency of the HPGe detector was both measured and calculated by the program Gamma Vision (Table 6), while the measurement and calculation results almost coincided.

The measurement uncertainty for the HPGe spectrometer was determined as through the minimum detectable activity (MDA) during counting time of 60,000 s (Table 7).

As far as for the scintillation spectrometer is concerned, the total measurement uncertainty was calculated as the sum of the statistical (ΔA_{XStat}) and systematical (ΔA_{XSys}) errors of activity concentration A_X for each radionuclide:

$$\Delta A_X = \sqrt{\Delta A_{XStat}^2 + \Delta A_{XSys}^2} \quad (7)$$

The systematic error of the given spectrometer was accepted as 5% of A_X .

The random (statistical) error (at 95% confidential level) was determined as:

$$\Delta A_{XStat} = \sqrt{\sum_{i=1}^n \left(\frac{\partial A}{\partial (S_i - F_i)} \right)^2 \cdot (\Delta S_i^2 + \Delta F_i^2)} \quad (8)$$

where ΔS_i and ΔF_i are absolute values of the statistical uncertainty of the measured and background count rates in the interval i , respectively; the value of the derivative $\frac{\partial A}{\partial (S_i - F_i)}$ was calculated numerically by solving the system of equations for the count rate S_i changed by 1%.

Table 5
Experimental plan (synthetic mixtures).

Mixture no.	²²⁶ Ra	²³² Th	⁴⁰ K
1	–	+	–
2	–	+	+
3	–	–	+
4	+	+	–
5	+	–	–
6	+	–	+
7	+	+	+
8	–	–	–

Table 6
Efficiency of the HPGe detector.

Energy (keV)	Measured efficiency	Efficiency calculated by gamma vision
186	2.320E-02	2.390E-02
242	1.860E-02	1.810E-02
295	1.512E-02	1.476E-02
351	1.275E-02	1.236E-02
609	7.102E-03	7.383E-03
1120	4.453E-03	4.406E-03
1764	3.189E-03	3.312E-03
2204	2.745E-03	2.665E-03
2447	2.414E-03	2.419E-03

Table 7
MDA (HPGe detector)

Nuclide	Energy (keV)	MDA (Bq)
²²⁶ Ra (²¹⁴ Bi)	609	0.72
²³² Th (²²⁸ Ac)	911	0.95
⁴⁰ K	1460	4.07

8. Preparation of the samples

The mixtures were prepared from four raw materials: (a) phosphate ore (with elevated ²²⁶Ra content); (b) powder prepared by mixing accurate amounts of ThO₂ (99.9% purity, $3,565 \pm 15$ Bq/g) with pure CaCO₃ and subsequent homogenization, (c) powder of KCl and (d) wheat powder as a “blank” material, which was used for dilution of the activity. The preparation procedure was similar to that described recently (Alfassi and Lavi, 2005; Lavi and Alfassi, 2005). To achieve the targeted activities, the raw materials in the needed proportions were thoroughly mixed and homogenized.

Regarding the preparation of the ²³²Th standard and possible concern about the absence of its equilibrium with other thorium isotopes, it has to be emphasized that ²³²Th is in equilibrium with ²²⁸Th, since the half-life of ²²⁸Th is only 1.9 years. The ²²⁸Th is also in equilibrium with ²²⁴Th, since it has a half-life of 1.03 s only. All the other alpha emitters in the thorium chain are even shorter lived.

In addition to the synthetic mixtures, real samples of building materials, such as hardened concrete, pumice and gypsum, have been tested with both types of detectors.

The prepared powder samples were transferred into Marinelli beakers of 1 l.

9. Results and discussion

Comparative measurement results deduced from the calibrated HPGe and NaI(Tl) detectors are shown in Fig. 3. As can be seen, the measurement results obtained by means of NaI(Tl) detector coincide perfectly with the results obtained by means of HPGe detector. The results are presented in the form of radium equivalent activity concentration A_{Ra-eq} calculated by the formula suggested by Beretka and Mathew (1985):

$$A_{Ra-eq} = A_{Ra} + 1.43A_{Th} + 0.077A_K \quad (9)$$

where it is assumed that 1 Bq/kg of ²²⁶Ra, 0.7 Bq/kg of ²³²Th or 13 Bq/kg of ⁴⁰K produce the same γ -ray dose rate (Krisiuk et al., 1971; Strandén, 1976), while the error of radium equivalent activity concentration is calculated as

$$U_{Ra,eq} = \sqrt{\frac{(U_{Ra})^2 + (1.43U_{Th})^2 + (0.077U_K)^2}{6}} \quad (10)$$

where U_{Ra} , U_{Th} and U_K are the measurement errors of ²²⁶Ra, ²³²Th and ⁴⁰K activity concentrations, respectively.

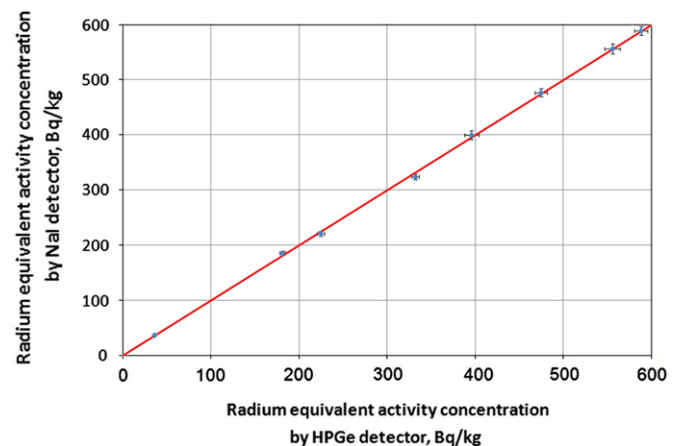


Fig. 3. Radium equivalent activity concentrations of the synthesized mixtures measured by NaI(Tl) detector vs. those measured by HPGe detector.

In addition to the synthetic mixtures, the samples of building materials, such as hardened concrete, pumice and gypsum, have been tested with both types of detectors. The results are shown in Table 8. In these tests the effect of counting time in NaI(Tl) spectrometer has been evaluated as well. The results show that by varying the counting time, the required accuracy, less than 15% error, can be obtained easily.

As can be seen, the measurement results by means of NaI(Tl) detector coincide very well with the results by means of HPGe detector, much better than in the work by Chiozzi et al. (2000). It has to be mentioned that the difference between the results obtained on NaI(Tl) and HPGe detectors for all the samples tested is significantly lower than the measurement error, which is calculated by the program of spectrum processing using matrix method with 12 energy intervals. It is explained by the fact that both detectors have been calibrated with the same set of the calibration sources; it means both detectors have the same systematic error, which is a part of the total error reported in Table 8.

Table 8 shows that the results obtained in the NaI(Tl) system are similar to those obtained with the HPGe spectrometer, mostly within the uncertainty range. This comparison proves that scintillation spectrometers equipped with a special software aimed to compensate for the lower spectral resolution of NaI(Tl) detectors can be successfully used for the radiation control of mass construction products.

In our opinion, using such software in scintillation spectrometers is obligatory for accurate quantitative measurement of NORM in building materials. However, this should be only a part of the standard requirements. The measurement system based on scintillation detector (as well as HPGe detector, by the way) should meet a number of basic requirements to energy calibration, uncertainties of efficiency calibration, protection from background radiation, maximum limit of detection and limit of quantification, such as those set in the standard published recently (SI 5098, 2009).

10. Conclusions

In many applications, such as the mass control of radioactive impurities in building materials, it is not practical to use a cooled high-purity germanium spectrometer. The improved efficiency, for the same size detector, and the lower cost of a NaI(Tl) detector can be traded-off against the better resolution of HPGe detector. At the same time, there is a need to compensate for the lower spectral resolution of NaI(Tl) detectors.

The NaI(Tl) spectrometer equipped with the special software based on the matrix method of least squares provides an accurate quantitative analysis of the radionuclides in (a) the synthetic compositions with activity concentrations varying in a wide range, from 1/5 to 5 times median activity concentrations of the

Table 8

Activity concentrations measured by γ -ray systems with HPGe and NaI(Tl) detectors and measurement errors (conf. level=95%).

Sample	Mass (kg)	Detector	Count. time (s)	Activity concentrations (Bq/kg)		
				^{226}Ra	^{232}Th	^{40}K
1	0.880	HPGe	78334	< 0.4	268.9 ± 14.0	146.5 ± 7.8
		NaI(Tl)	14400	< 2.4	272.0 ± 13.9	140.2 ± 13.0
2	0.864	HPGe	92605	6.0 ± 0.4	266.3 ± 14.0	2196 ± 109
		NaI(Tl)	14400	5.3 ± 2.7	270.4 ± 13.9	2137 ± 110
3	0.892	HPGe	52182	7.1 ± 0.4	8.5 ± 0.5	2102 ± 105
		NaI(Tl)	14400	7.0 ± 1.5	7.9 ± 1.6	2168 ± 111
4	0.944	HPGe	67577	166.4 ± 8.6	207.6 ± 8.6	148.0 ± 8.1
		NaI(Tl)	14400	165.8 ± 8.7	209.1 ± 10.8	153.9 ± 19.4
5	0.872	HPGe	53155	195.1 ± 10.3	13.5 ± 0.8	129.7 ± 7.3
		NaI(Tl)	14400	191.6 ± 9.8	13.5 ± 1.9	129.7 ± 17.0
6	1.010	HPGe	267814	178.7 ± 9.0	11.4 ± 0.6	1781 ± 89
		NaI(Tl)	14400	174.9 ± 9.0	10.0 ± 1.7	1747 ± 90
7	0.978	HPGe	86232	159.5 ± 7.9	202.8 ± 10.1	1803 ± 92
		NaI(Tl)	14400	156.4 ± 8.3	205.6 ± 10.7	1800 ± 94
8	0.888	HPGe	70772	7.5 ± 0.4	11.8 ± 0.6	151.7 ± 12.4
		NaI(Tl)	14400	7.6 ± 1.2	12.2 ± 1.4	149.9 ± 13.0
Pumice	1.269	HPGe	63320	49.3 ± 3.0	44.5 ± 2.2	731.7 ± 36.6
Pumice	1.276	NaI(Tl)	50000	49.8 ± 1.1	49.6 ± 1.1	735.7 ± 9.6
		HPGe	63320	48.5 ± 2.8	45.4 ± 2.4	801.0 ± 46.4
Pumice	1.042	NaI(Tl)	7200	49.0 ± 3.2	47.6 ± 3.1	777.3 ± 43.9
		HPGe	63320	53.1 ± 3.2	47.7 ± 2.6	880.9 ± 52.6
Gypsum	1.008	NaI(Tl)	7200	54.6 ± 3.6	53.0 ± 3.6	914.2 ± 51.8
		HPGe	254886	9.3 ± 0.5	0.8 ± 0.1	9.3 ± 1.1
Concrete	2.410 ^a	NaI(Tl)	50000	8.9 ± 0.8	< 0.8	7.2 ± 6.1
		HPGe	60000	23.3 ± 1.3	4.0 ± 0.2	66.7 ± 4.3
Concrete	2.390 ^a	NaI(Tl)	7200	22.2 ± 1.4	3.8 ± 0.8	64.3 ± 7.6
		HPGe	60000	28.0 ± 1.5	9.2 ± 0.5	69.4 ± 4.5
Concrete	2.376 ^a	NaI(Tl)	7200	26.0 ± 1.7	9.6 ± 1.0	62.9 ± 8.0
		HPGe	14320	34.3 ± 2.0	15.5 ± 0.9	71.1 ± 7.0
Concrete	1.750	NaI(Tl)	7200	31.9 ± 1.9	15.4 ± 1.3	56.7 ± 8.2
		HPGe	60000	22.0 ± 1.2	4.1 ± 0.2	67.9 ± 4.7
Concrete	1.900	NaI(Tl)	7200	23.2 ± 1.6	4.1 ± 1.0	63.1 ± 9.0
		HPGe	60000	23.8 ± 1.3	8.9 ± 0.5	62.0 ± 4.4
Concrete	1.900	NaI(Tl)	7200	22.1 ± 1.5	8.9 ± 1.1	60.6 ± 8.6
		HPGe	60000	29.9 ± 1.6	15.0 ± 0.8	66.3 ± 4.6
Concrete	1.900	NaI(Tl)	7200	26.4 ± 1.8	14.0 ± 1.3	57.5 ± 8.9

^a The relatively high density of the sample is due to the sampling procedure: the sample was cast directly in the Marinelli beaker and tested as a monolith material, without crushing/pulverizing of the hardened mass.

natural radionuclides available in the earth crust and (b) the samples of popular building materials, such as concrete, pumice and gypsum, while the density of the tested samples changed in a wide range: from 860 up to 2,410 kg/m³. The main feature of the applied software is that the entire spectrum is represented as a sum of the functions of the response to the individual radiation components, instead of the treatment of the peaks of the full energy absorption.

The results obtained in the NaI(Tl) system are similar to those obtained with the HPGe spectrometer, mostly within the uncertainty range. This comparison shows that scintillation spectrometers equipped with a special software aimed to compensate for the lower spectral resolution of NaI(Tl) detectors can be successfully used for the radiation control of mass construction products.

Acknowledgments

The authors are thankful to Dr. Uzi German and Mr. Shmuel Levinson for the valuable comments and suggestions.

References

- Alfassi, Z.B., Lavi, N., 2005. The dependence of the counting efficiency of Marinelli beakers for environmental samples on the density of the samples. *Appl. Radiat. Isot.* 63, 87–92.
- Antropov, S.Y., Ermilov, A.P., Ermilov, S.A., Komarov, N.A., Krochin, I.I., Nikolaev, A.V., 2003. The Method of Measurement of Radionuclides' Activity Using Scintillation Gamma-Spectrometer with "Progress" Software, Moscow: National Research Institute for Physicotechnical and Radio Engineering Measurements (in Russian).
- Antropov, S.Y., 2004. Progress-5. Adjustment and Programming of Spectrometric Devices. Amplituda Research Center, Moscow (in Russian).
- Beretka, J., Mathew, P.J., 1985. Radioactivity of Australian building materials, industrial wastes and by-products. *Health Phys.* 48, 87–95.
- Chiozzi, P., De Felice, P., Fazio, A., Pasquale, V., Verdoya, M., 2000. Laboratory application of NaI(Tl) γ -ray spectrometry to studies of natural radioactivity in geophysics. *Appl. Radiat. Isot.* 53, 127–132.
- Gilmore, G.R., 2008. *Practical Gamma-Ray Spectrometry*. John Wiley & Sons, Ltd., Chichester.
- GOST 30108, 1994. Building materials and elements. Determination of specific activity of natural radioactive nuclei, Moscow: Gosstandart, (in Russian).
- Guide, S.T., 2005. *The Radioactivity of Building Materials and Ash*. Helsinki, Finland: STUK.
- IAEA, 1987. Preparation of γ -ray Spectrometry Reference Materials RGU-1, RTh-1 and RKG-1, Report-IAEA/RL/148. Vienna: IAEA.
- IAEA, 2011. *Safeguards Techniques and Equipment*. Vienna: IAEA, International Nuclear Verification Series.
- Krisiuk, E.M., Tarasov, S.I., Shamov, V.P., Shalak, E.P., Lisachenko, N.I., Gomeltsky, L.G., 1971. A Study on Radioactivity in Building Materials. Leningrad: Leningrad Research Institute for Radiation Hygiene, USSR.
- Lavi, N., Alfassi, Z.B., 2005. Development of Marinelli beaker standards containing thorium oxide and application for measurements of radioactive environmental samples. *Radiation Measurements* 39, 15–19.
- Meng, L.J., Ramsden, D., 2000. An inter-comparison of three spectral-deconvolution algorithms. *IEEE Transactions on Nuclear Science* 47, 1329–1336.
- Perez-Andujar, A., Pibida, L., 2004. Performance of CdTe, HPGe and NaI(Tl) detectors for radioactivity measurements. *Appl. Radiat. Isot.* 60, 41–47.
- RP-112, 1999. Radiological Protection Principles Concerning the Natural Radioactivity of Building Materials, Radiation Protection Report RP-112, Luxembourg: European Commission.
- Rybach, L., 1988. Determination of the heat production rate. In: Rybach, R., Haenel, R., Rybach, L., Stegena, L. (Eds.), *Handbook of Terrestrial Heat-Flow Density Determination*. Kluwer Academic Publishers, Dordrecht, pp. 125–142.
- SI 5098, 2009. Content of Natural Radionuclides in Building Products. The Standards Institution of Israel: Tel-Aviv.
- Stranden, E., 1976. Some aspects on radioactivity of building materials. *Phys. Norv.* 8, 167–177.
- UNSCEAR, 2000. Sources and Effects of Ionizing Radiation. Annex B: Exposures from Natural Radiation Sources, New York: United Nations Publications.